Oisin Argand, 2nd Year

54/60 = 90% Excellent

Sep 25

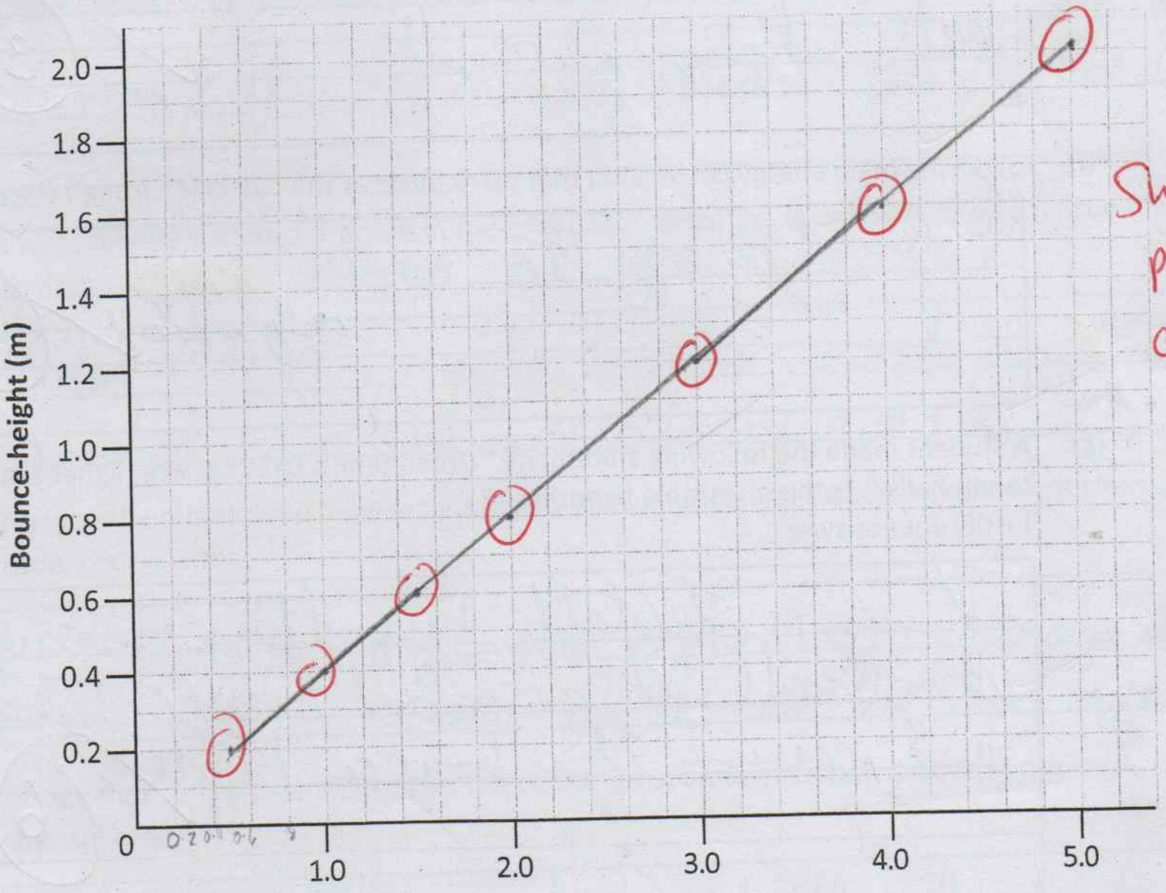
Question 12

(45 marks)

A group of students investigated how high a ball bounces after it is dropped.
They allowed a ball to fall from a number of different drop-heights and measured how high the ball bounced each time (the bounce-height).
Their results are shown in the table below.

Drop-height (m)	**Bounce-height (m)**
0.5	0.2
1.0	0.4
1.5	0.6
2.0	0.8
3.0	1.2
4.0	1.6
5.0	2.0

(a) In the space below, present this information using a suitable graph or chart.
(Your graph or chart should allow the viewer to read the results of the investigation and to see any pattern.)



Show points clearly.

18

(b) If the ball was dropped from a height of 10 m, how high would you expect it to bounce?

4 meters

3

(c) Which is easier to measure accurately, the drop-height of the ball or its bounce-height? Explain your answer.

Drop height because the ball can be there for as long as you need to measure it

6

(d) Which is easier to measure accurately, 0.2 m or 1.6 m? Explain your answer.

0.2 meters because you can use smaller, more accurate ~~devices~~ measuring tools and taller ones could sway more

(e) Outline one safety precaution which the students should take when carrying out this investigation.

Tie back hair so as not to hit it off the ball, changing results

3

(f) State the main energy conversion that takes place as the ball falls through the air.

Potential energy to kinetic energy

really important

6

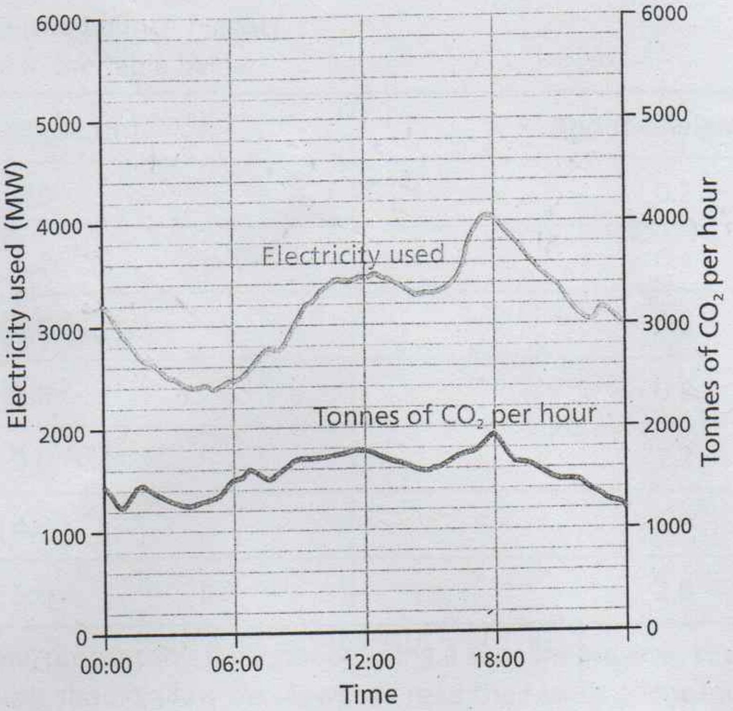
(g) A student made the following statement: "Green tennis balls bounce higher than orange tennis balls." Is this a testable hypothesis?
Justify your answer.

It is a Testable hypothesis because they could get green and orange tennis balls and measure their bounce height

6

Question 3 **15 marks**

The graph below shows the relationship between electricity consumption and carbon dioxide production in Ireland over a 24-hour period.



(a) At what time(s) is there a peak in demand for electricity? Suggest a reason for this peak.

Around 18:00, maybe because people are cooking dinner

3

(b) At what time(s) is there a drop-off in the production of carbon dioxide? Suggest a reason for this drop-off.

After 18:00, maybe because people are finished working and relaxing more

3

(c) Describe the relationship between electricity consumption and carbon dioxide production. Justify your answer.

Electricity consumption often uses fossil fuels which create CO_2

use the word increase

3

(d) Carbon dioxide is a greenhouse gas. List two other sources of carbon dioxide in the atmosphere.

Volcanic eruptions and animals breathing it out

use respiration

3